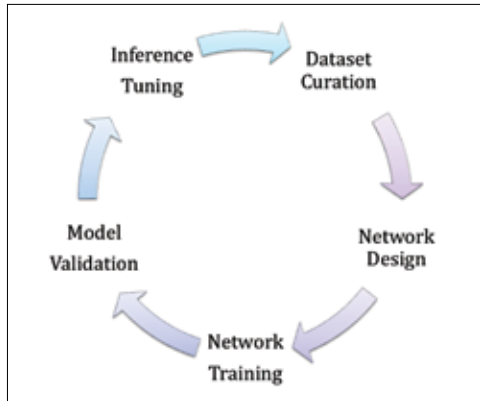


What is deep learning

From automated or driverless cars to efficient speech recognition, deep learning has transcended to being the technology that makes everything we considered impossible and insurmountable a very real possibility. Deep Learning is a subset of Machine Learning (ML), which in itself is a segment of Artificial Intelligence (AI). The average training cycle of AI consists of three distinct steps - training, inference and validation. Training is essentially giving a neural network input. Once the AI is trained utilising the input, you can ask it to automatically generate output or even give it partial input, where it fills in the rest. Training usually takes quite long and is very resource heavy. Towards the end of training, the validation stage sets in which is utilised to get rid of biases by providing the AI model additional datasets. Inference refers to the process of deducing information or results by applying your model to brand new data. It is essentially applying the trained model to unseen data to analyse its performance.



Stages of AI training - training, validation and inference, followed by dataset curation and the ultimate network design.

Deep learning is carried out through large Artificial Neural Networks (ANNs), usually over 100 layers of neurons. Training these neural networks mandates a lot of labelled data along with massive computing power. The applications of deep learning are manifold and extensive, therefore it has ended up being a hot topic of discussion.

Being part of the extended family of ML, it is concerned with automated devices learning data representations rather than task-specific algorithms. This complex sub-set of ML is inspired by the functioning of the human brain and its understanding of information. Deep learning expects to accomplish this gargantuan task by mimicking the activity in the layers of neurons present in the neocortex, which is the worm-like 80% of the brain where thinking occurs.